SOMYA RANJAN TRIPATHI

Software Development Engineer (SDE) | Backend Development | Neural Modeling

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PROFILE

Software Development Engineer specializing in high-performance **backend systems** and **end-to-end AI pipeline development**. Proven capability in architecting **data-driven infrastructure** and deploying **predictive machine learning models** for **complex analytical challenges**. Adept at building **scalable cloud architectures** and orchestrating **agentic AI workflows** for optimization.

EDUCATION

Babu Banarasi Das (BBD) University — Lucknow, Uttar Pradesh
Bachelor of Technology in Computer Software Engineering | CGPA: 7.41/10.0 — August 2023-September 2027

Delhi Public School (Shaheed Path) — Lucknow, Uttar Pradesh
Class XII (PCM & CS): 63.40% | Class X: 91.83% — July 2018-March 2023

EXPERIENCE

GRASTech | Internship — On-site, Lucknow
AWS Cloud Intern — June 2026-August 2026

- Architected scalable AWS infrastructure (EC2, S3, VPC, Lambda) and implemented foundational DevOps practices utilizing CI/CD, auto-scaling, and IaC.
- Configured secure network routing and automated deployment workflows to ensure highly reliable and scalable enterprise cloud operations.

YBI Foundation | Internship — Remote, Delhi
Artificial Intelligence & Machine Learning Intern — September 2024-November 2024

- Developed and trained the "Valley Prediction Model," utilizing advanced regression analysis on 120,000+ data points to extract structured insights from complex datasets.
- Engineered automated Python data preprocessing pipelines, optimizing dataset ingestion and significantly enhancing overall model training efficiency.

ACADEMIC PROJECTS

LekhaBit Platform: A Scalable SaaS Production Initiative — August 2026 – October 2026

- Spearheaded Java backend development for a collaborative SaaS platform supporting 10,000+ concurrent users.
- Optimized REST API endpoints and data routing to ensure high-performance synchronization and open-source integration.

Campus-Sync: Cloud-Native Synchronization Platform — June 2026 - July 2026

- Engineered a containerized FastAPI and React.js backend via Docker on AWS, reducing data latency by 40%.
- Automated cloud deployment pipelines to maintain 99.9% uptime across all application environments.

Ambaram Sentinel: AI Weather Prediction Model on Extreme Weather Events — July 2025 - March 2026

- Architected an AI-driven predictive system processing real-time meteorological data ingestion pipelines at 24+ MB/s.
- Implemented Python regression models to parse high-volume datasets for accurate extreme weather forecasting.

TECHNICAL SKILLS

- **Languages/Databases:** Java, Python, C++, JavaScript, TypeScript, MySQL, PostgreSQL, MongoDB, HTML5, CSS.
- **Web/App Development:** Spring Boot, Flutter, Next, React, Node, Django, Flask, FastAPI, REST APIs, Streamlit.
- **DevOps/Others:** Docker, AWS, Kubernetes, Linux, CI/CD, GitHub Actions, PyTorch, LangChain, Hugging Face.

CERTIFICATIONS

- **Technical Training Program| Learnovate Enterprises** — September 2026
- **Python Backend Developer | FreeCodeCamp** — May 2026
- **Relational Database | FreeCodeCamp** — May 2026
- **JavaScript Developer | FreeCodeCamp** — May 2026
- **Foundation: Monitoring Production Agents | Langchain Academy** — May 2026